

Cryptic phenotypic variation emerges rapidly during the adaptive evolution of a carbapenemase

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Interactions among beneficial mutations (that is, epistasis) are often strong enough to direct adaptation through alternative mutational paths. Although alternative solutions should display similar fitness under the primary selective conditions, their properties across secondary environments may differ widely. The extent to which these cryptic differences are to be expected is largely unknown, despite their importance—for example, in identifying exploitable collateral sensitivities among mutations conferring antibiotic resistance. Here we use directed evolution to characterize the diversity of mutational paths through which the prevalent carbapenemase *Klebsiella pneumoniae* carbapenemase-2 can evolve high activity against the clinically relevant antibiotic ceftazidime, an initially poor substrate. We identified 40 different substitutions—including many that are common in clinical settings—spread among 18 different mutational trajectories. Initial mutations determined four major groups into which the trajectories can be classified, a signature of strong epistasis. Despite similar final ceftazidime resistance, groups diverged markedly across multiple phenotypic dimensions, from molecular traits, such as in-cell stability and catalytic efficiency, to macroscopic traits, such as growth rate and activity against other β -lactam antibiotics. Our results indicate that cryptic yet consequential phenotypic differences can accumulate rapidly under strong selection, unpredictably shaping the long-term success of resistance enzymes in their journey across hosts and environments.

Adaptation inevitably reduces genetic diversity. Alleles reaching fixation reduce variation not only at their locus, but also at all other regions not separated by recombination¹ (that is, a selective sweep^{2,3}). Early models assumed that beneficial mutations were rare, so sweeps would typically have a single origin and drastically reduce diversity (that is, a hard sweep)^{4,5}. Recent work, however, shows that adapting populations are often large enough to harbour multiple genotypes representing alternative solutions to the same challenge (that is, a soft sweep)^{6–9}. Whether this genetic variation can have evolutionary

consequences beyond the local selective conditions is crucial to many problems in ecology and evolution. Examples include understanding biodiversity patterns along environmental gradients^{10,11}, the tempo and mode of speciation^{12,13} and the capacity of populations to adapt to new challenges^{14,15}, which is crucial both in conservation and for pest and pathogen control¹⁶.

This issue is particularly relevant in the fight against antibiotic resistance evolution, a major public health threat¹⁷. Next-generation sequencing has revealed that bacterial populations that have adapted

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to antibiotics often comprise multiple clones with different resistance determinants^{6,18–20}. Although they are presumably equivalent in fitness under the primary selective conditions, these alternatives may display distinct liabilities and strengths across other relevant conditions. This possibility has direct relevance for strategies to prevent or eliminate antibiotic-resistant variants^{21–24}. One such approach is prioritizing antibiotics for which resistance entails a substantial fitness cost (for example, reduced growth, transmission or virulence, or increased sensitivity to the host's defences²⁵). Another, more recently advocated strategy is the use of drug combination or cycling regimens that exploit trade-offs among antibiotics (that is, collateral sensitivity, the observation that resistance to a given drug often induces hypersusceptibility to other drugs^{26,27}).

Although promising, the clinical success of these strategies hinges on the assumption that evolutionary outcomes are largely predictable, an assumption that is far from guaranteed under real-world conditions^{28–30}. At least two important cautionary points can be raised. First, resistance mutations are rarely functionally identical, being better depicted as a pool of different classes with varying frequencies and phenotypic profiles^{28,31–33}. Consequently, the behaviour of a typical mutant may not represent the full spectrum of possibilities. Second, changes in population genetic parameters can alter which mutations appear as typical: variations in population size, mutation bias and selection strength can lead to different subsets of mutations fixing during adaptation^{6,28,34–37}.

A second, less appreciated cautionary point concerns epistasis (that is, non-additive interactions among mutations). Epistasis is common among beneficial mutations, reflecting the many nonlinearities in the genotype-to-fitness map that arise at structural, metabolic and regulatory levels^{38–41}. Of particular relevance is strong epistasis, which can reverse the direction of fitness effects when mutations are combined (that is, sign epistasis)⁴². This poses two challenges for exploiting trade-offs in applied settings. First, since new mutations may interact idiosyncratically with previous ones, trade-offs observed in a typical mutant may not persist further along an adaptive pathway, making initial snapshot measurements of collateral sensitivities unreliable²⁸. Second, strong epistasis renders mutational paths highly contingent on which mutations happen to occur first⁴³. As a result, even within a single population, lineages may rapidly diverge across multiple traits, making it less likely that a typical mutant accurately represents how populations behave in secondary environments.

Antibiotic resistance enzymes are an ideal model with which to explore how epistasis can drive phenotypic divergence, both across and within mutational paths. First, they are major contributors to clinical resistance, posing substantial concern due to their potential for cross-species spread^{44,45}. Second, strong epistasis is pervasive among beneficial mutations within enzymes, as different residues often interact either through direct physical contact or indirectly via altering stability and folding. Indeed, much of our understanding of intramolecular epistasis stems from studies of the β -lactamase enzyme TEM-1 (refs. 41–43,46–50). Third, resistance enzymes often show substrate promiscuity across multiple antibiotics within the same class, and the clinical evolution of increased activity against non-preferred substrates is well documented^{51,52}. As a result, these enzymes afford great opportunity to explore how adaptation to non-preferred substrates can follow phenotypically distinct paths.

We used directed evolution to assess the extent to which epistasis can drive rapid, multi-trait divergence in the clinically relevant carbapenemase *Klebsiella pneumoniae* carbapenemase-2 (KPC-2)⁵³. This class A serine β -lactamase was originally named for its ability to hydrolyse carbapenems—last-resort drugs for multidrug-resistant infections⁵⁴. However, KPC-2 and its variants also act on a wide range of other β -lactams, including penicillins, monobactams and cephalosporins^{55,56}. Their broad-spectrum phenotype, global spread and high prevalence in Gram-negative bacteria have made KPC carbapenemases a top priority

for epidemiological surveillance^{53,57}. KPC-2 also shows high evolutionary potential⁵⁸, with over 200 allelic variants reported in less than three decades since it was first described (<http://bldb.eu>; ref. 59). These variants have mostly diverged through point mutations, in many cases resulting in substantial alterations of their activity spectrum^{58,60}.

One of the few, yet relevant, compounds for which KPC-2 shows poor activity is ceftazidime (CAZ)⁶¹, an important β -lactam cephalosporin that is commonly used to treat Gram-negative and some Gram-positive bacteria⁶². Leveraging this weakness, CAZ is used in combination with β -lactamase inhibitors, such as avibactam, to treat KPC-bearing pathogens^{63–65}. However, despite the fact that CAZ–avibactam combination was only recently introduced (2015 in the USA and 2016 in Europe), numerous reports document the emergence of resistant mutants, often via mutations that reshape active-site loops to increase activity against CAZ^{58,60,66,67}. Inspired by this natural case study, here we used in-vitro-directed evolution to characterize the diversity of mutational paths by which KPC-2 can acquire increased resistance to CAZ, focusing particularly on how readily these paths can diverge not only genetically but also across multiple phenotypic traits.

Results

Initial CAZ resistance arises via clinically common variants

To explore the potential of KPC-2 to evolve high activity against CAZ, we cloned a wild-type *bla*_{KPC-2} gene into a high-copy-number plasmid in *Escherichia coli* and then used error-prone PCR to create 11 libraries with $\sim 10^3$ – 10^4 mutants. After overnight growth, we screened for CAZ resistance phenotypes across a gradient of doubling concentrations of CAZ, starting from the wild type's minimum inhibitory concentration (MIC; 16 $\mu\text{g ml}^{-1}$; Methods). From each library, we selected two colonies from the highest CAZ concentration sustaining growth, which in this first round amounted to a 16- to 32-fold MIC increase (256–512 $\mu\text{g ml}^{-1}$). Sanger sequencing of these initial 22 colonies revealed 38 single-nucleotide polymorphisms in the *bla*_{KPC-2} gene (1.54 ± 0.66 per isolate; values represent mean $\pm$ s.d.), of which 15 were unique. These mutations translated into 34 amino acid substitutions (13 unique), arranged into nine genotypes (Supplementary Table 1 and Fig. 1a).

Next, we classified the first-round mutants according to the mutations most likely to be driving the MIC increases. Clinical CAZ-resistant variants typically harbour mutations in three structural hotspots^{58,61}: the Ω -loop, forming the catalytic pocket floor; loop 240, forming the active site's right wall; and loop 270, which changes conformation upon CAZ binding. Each of our initial 22 mutants carried mutations encoding changes to either the Ω -loop or loop 270, allowing us to define four classes based on previously characterized resistance-associated positions. The first class involved position D179, located in the Ω -loop. This position was the most commonly altered among the initial 22 colonies (13/22 (59%); Supplementary Table 1) and the change was present in over half of the initial genotypes (5/9; Fig. 1a). This prevalence may be linked to it being the position where a single amino acid substitution confers the highest MIC increase (D179Y; 32-fold), although mutational biases may also contribute⁶⁸. Interestingly, this position is also the most commonly hit among CAZ-resistant clinical variants⁵⁸. We also observed substitutions at other positions within the Ω -loop, specifically L169 and S171 (2/22 (9%) each; Supplementary Table 1), which we considered to be different classes with potentially different evolutionary prospects. The final class comprises position H274, in loop 270, the second most commonly altered among our mutants (5/22 (23%); Supplementary Table 1). Of note, KPC-2 with the single substitution H274Y is known as KPC-3 in the literature⁶⁹ and is the most prevalent variant overall after KPC-2 (ref. 70).

KPC-2 can follow alternative routes towards CAZ resistance

Despite starting with 22 lines, the first round produced only nine different genotypes, falling into four classes based on probable CAZ resistance mutations. Would these variants lead to markedly different

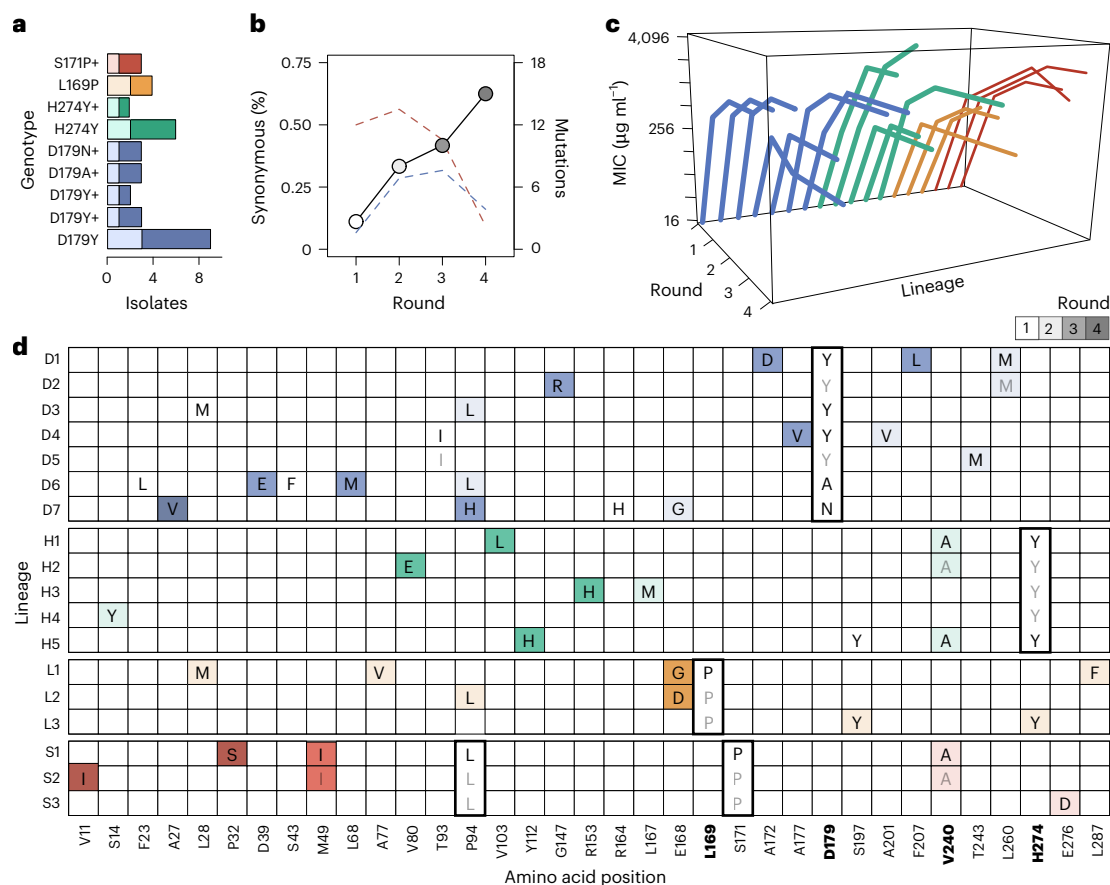


Fig. 1 | Mutational pathways recovered during KPC-2 directed evolution for increased CAZ resistance. **a**, Abundances of the nine unique KPC-2 variants retrieved among the initial 22 colonies isolated in round 1. They have been grouped into four categories (as indicated by a difference in colour), according to the location of the amino acid substitution at one of four positions previously shown to be associated with CAZ resistance. A plus sign indicates additional accessory mutations in the gene encoding KPC-2 (Supplementary Table 1). Darker shades represent the total abundance across replicates, whereas lighter shades indicate independent occurrences. **b**, Proportion of synonymous mutations across rounds (solid black line with circles; darker grey shades represent later rounds). The secondary y axis displays the total number of synonymous (blue

dashed line) and non-synonymous mutations (red dashed line). **c**, Diversity of CAZ resistance trajectories across rounds and lineages, grouped and coloured as in **a**. An overlaid, two-dimensional representation of the mutational trajectories is shown in Extended Data Fig. 1. Note the y-axis is on a base-2 logarithmic scale. **d**, Horizontal alignment of the 18 lineages studied, highlighting only the positions with alterations. The lineages are arranged, grouped and coloured as in **a**, with darker shades representing later rounds. Duplicated lines are shown in grey up to the round in which they were actually founded. Alterations encoded by presumed driver mutations are highlighted with a thick outline to underscore that they are the basis for defining each group. The bold labels on the x axis mark the four substitutions used to construct the adaptive landscape shown in Fig. 2.

evolutionary outcomes? To better explore less common trajectories, we conducted a second round of directed evolution not only with the initial nine genotypes but also with six duplicates chosen from across the four groups. Specifically, we added one line from each of the D179 and S171 groups (lines D5 and S3, respectively; Supplementary Table 2) and two lines from each of the H274Y and L169 groups (H3, H4, L2 and L3; Supplementary Table 2). These latter groups were favoured due to their lower MIC gains (16-fold), which are potentially a hallmark of suboptimal paths⁵⁰. The resulting 15 lines underwent a new round of mutagenesis and screening. Whereas in round one we observed consistently large MIC increases (16- and 32-fold), MIC changes in round two were much smaller and heterogeneous (from 0.5- to eightfold) (Supplementary Table 2). This heterogeneity suggests that the lines had already begun to diverge substantially, with most still undergoing adaptation (10/15), a few reaching a plateau (4/15 (that is, lines D5, H3, H4 and L1); Supplementary Table 2) and one even declining in MIC, probably due to the random sampling of a deleterious mutation (that encoding an A201V substitution; line D4; Supplementary Table 2).

For the third round of directed evolution, we considered three criteria. First, we included the ten lines still adapting thus far. Second, we duplicated two lines with the highest MICs to explore alternative

routes to maximum CAZ resistance (D1 and H1, with MICs of 1,024 and 2,048 $\mu\text{g ml}^{-1}$, respectively (a 64- and 128-fold increase compared with the wild type)). Third, we included three out of the four lines that plateaued in the previous round, plus the line that declined in MIC. These were included to investigate what drove their apparent halt in adaptation: was it due to MIC-increasing mutations being rare or entirely absent? In total, we proceeded with 16 lines for a third round of mutagenesis and screening. Unlike previous rounds, here most lines showed no further adaptation (14 out of 16) and the only two lines that did exhibited only a modest twofold increase in MIC (H2 and S1; Supplementary Table 2).

For the fourth and final round of mutagenesis, we chose a total of nine lines: duplicates of the two still adapting (although for H2 no further mutations appeared so the duplicate is not included in Supplementary Table 2) and five lines across groups that, despite having already plateaued, displayed either some of the highest MICs (D6, D7 and H5; Supplementary Table 2) or some of the lowest MICs (D4 and L1; Supplementary Table 2). We included these stalled lines to explore whether the non-beneficial mutations acquired in the previous round could still act as stepping stones towards further adaptation⁷¹. None of the nine lines showed additional MIC increases (Supplementary Table 2).

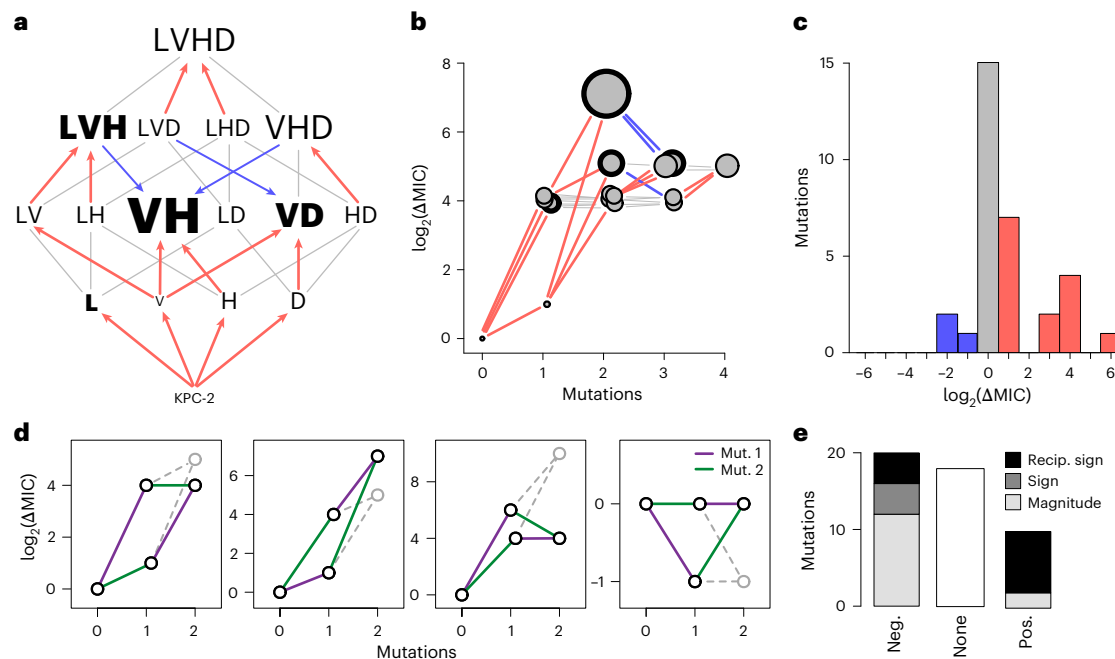


Fig. 2 | Adaptive landscape for all combinations of four key beneficial mutations during evolution towards CAZ resistance. **a**, Schematic of the accessibility of different mutational trajectories starting from the ancestral KPC-2 (bottom node). Beneficial alterations are depicted with red arrows, neutral ones with grey lines and deleterious ones with blue arrows. The node labels correspond to the initial letter of each substitution (that is, V240A, L169P, H274Y and D179Y). Node size is proportional to the MIC. Bold labels indicate the most likely outcomes under directed evolution, assuming that only beneficial mutations are accessible and back mutations are rare (note that these protein variants may only partially overlap with local peaks—typically defined as those with no fitter neighbours). **b**, Side view of the adaptive landscape, colour and node-size conventions are the same as in panel **a**. Resistance rapidly plateaus after the incorporation of one or two mutations, with the global fitness peak reached through a combination of mutations encoding the substitutions

H274Y and V240A. Further addition of originally beneficial mutations to this background results in inevitable fitness decay (blue lines). **c**, Distribution of individual fitness effects (measured as changes in CAZ susceptibility), showing that most mutations are neutral or deleterious across backgrounds, despite all being beneficial in the ancestor—highlighting the strength and pervasiveness of epistasis in this system. **d**, Examples of the different types of pairwise epistasis observed in the landscape (see Extended Data Fig. 2 for all cases). The grey dashed lines represent the null expectation of pure additivity. The first two panels illustrate magnitude epistasis, where the double mutant deviates moderately above or below the additive expectation. The third and fourth panels exemplify sign epistasis, where the interaction reverses the direction of the single-mutant effects (reciprocal sign epistasis if both are reversed). **e**, Distribution of epistasis types across the entire landscape. Mut., mutation; Neg., negative; Pos., positive; Recip., reciprocal.

These experiments revealed three major takeaways. First, at least four distinct sites can harbour large-effect, first-step CAZ resistance substitutions (Fig. 1a). However, these sites are unevenly represented, with one accounting for 59% of first-step substitutions (D179), presumably because it enables the highest single-step MIC increase. Second, although the proportion of synonymous mutations showed no evidence of plateauing (Fig. 1b), MIC gains slowed almost in parallel across groups, with most lines plateauing by round three (Fig. 1c and Extended Data Fig. 1). This contrast implies an excess of non-synonymous mutations during the initial versus later stages of plateaus, potentially indicating that plateaus are not adaptively stagnant. Are these non-synonymous mutations mediating cryptic adaptation in other fitness-related traits, despite not affecting MICs?

The third takeaway is that first-step mutations directed adaptation through distinct sets of adaptive mutations, the signature of sign epistasis⁴³. Indeed, only six of the 35 sites featuring substitutions were hit in more than one group (17%; L28, P94, E168, S197, V240 and H274; Fig. 1d), with only one site hit in three groups (P94) and none in all. To directly assess the prevalence and strength of epistasis, we engineered the full combinatorial adaptive landscape of four key substitutions: three encoded by first-round driver mutations (L169P, H274Y and D179Y) and one encoded by the most impactful second-round mutation (V240A, which increases the MIC eightfold when added to H274Y) (Supplementary Table 2). The resulting landscape revealed strong and widespread epistasis, with multiple local peaks and most mutations becoming neutral or deleterious across

backgrounds—despite all being beneficial in the ancestor (Fig. 2 and Extended Data Fig. 2).

Notably, the strong epistasis-driven genetic divergence yielded only moderate differences in CAZ resistance (Fig. 1c). Indeed, only groups L169 and S171 showed significant differences in endpoint MICs ($P = 0.024$; t -test on $\log_2$ -transformed values). In other words, the different groups may be exploiting alternative molecular solutions, but these solutions converge to similar values of the major trait under selection. Although this convergence may reflect fundamental physicochemical limits to enzymatic activity, it is still possible that the divergent genotypes differ across other traits relevant to fitness elsewhere. Could this genetic divergence translate into markedly different properties beyond the original selective conditions?

Collateral sensitivities rapidly diverge during adaptation

Next, we examined the extent to which the CAZ-adapted lines displayed cross-resistance or cross-sensitivity to other β -lactam antibiotics, as cycling among β -lactams has been proposed as a way to curb resistance evolution^{72–74}. We chose three compounds commonly used to treat Gram-negative infections, with different degrees of chemical similarity to CAZ⁷⁵: cefotaxime (CTX), which belongs to the same subclass as CAZ (third-generation cephalosporins); cefoxitin (FOX), a second-generation cephalosporin; and imipenem (IMI), from the carbapenem class. Overall, CAZ-adapted lines tended to become more susceptible to the other antibiotics, although the fraction of lines increasing susceptibility varied considerably (92.9% for IMI, 78.6%

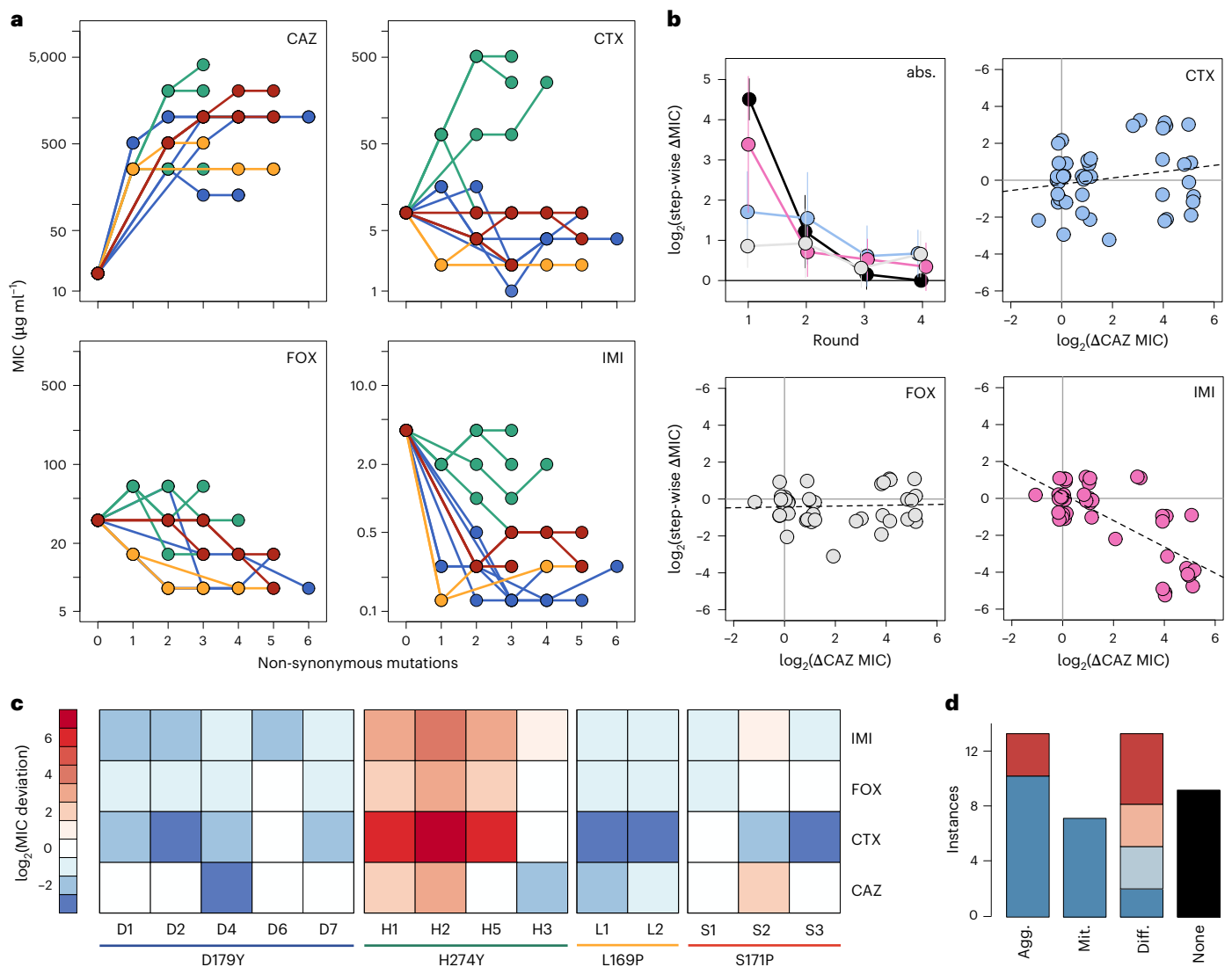


Fig. 3 | Collateral sensitivity towards other β -lactams rapidly diverges during CAZ resistance evolution. **a**, Resistance trajectories against the number of non-synonymous mutations accumulated across rounds for each lineage. The colours indicate the grouping based on the positions associated with CAZ resistance, where blue, green, yellow and red represent D179Y, H274Y, L169P and S171P, respectively. The panels show MIC measurements for CAZ, CTX, FOX and IMI, as labelled in the top right corner. A version of these panels with experimental rounds on the x axis is provided as Extended Data Fig. 1. Note the y-axis is on a base-10 logarithmic scale. **b**, Patterns of resistance across rounds. The top left panel shows the absolute change for each β -lactam (that is, the effect size). Points represent the arithmetic mean and error bars show standard deviation ($n = 14$). The colours denote averages for CAZ (black), CTX (cyan), FOX (grey) and IMI (pink). The remaining panels show correlations between individual MIC changes for CAZ and the other antibiotics, with dashed lines indicating the best fit from

linear regression. Note that each data point corresponds to a single evolutionary step within a lineage, enabling analysis of how resistance trade-offs evolve over time. abs., absolute. **c**, Variation in collateral sensitivity among and within lineages. The panels show resistance values for endpoint isolates are represented following a colour gradient, where blue represents increased sensitivity and red represents cross-resistance. Note that the values denote the relative variation compared with all lineages. **d**, Divergence in collateral sensitivity along mutational pathways. The values indicate the number of lineages for which the full path had aggravating (Agg.), mitigating (Mit.), different (Diff.) or no effects (None) on first-step mutant resistance. Blue indicates that the first step caused increased sensitivity and red indicates that it caused resistance. Lighter colours represent cases where the endpoint profile was neutral compared with the wild type.

for FOX and 57.1% for CTX; Fig. 3a). Furthermore, MIC trajectories for the other antibiotics showed frequent fluctuations after CAZ MICs had plateaued. This observation further suggests that non-synonymous mutations in later rounds, though not affecting CAZ MICs, may drive adaptation in other fitness components, manifesting in altered collateral sensitivity profiles.

Next, we examined how the magnitude of collateral effects changed through rounds. Since most high-impact CAZ resistance mutations emerged early on, the most pronounced collateral effects were expected to occur early as well^{76,77}. Our results confirmed this expectation, but also showed that although these effects tended to

decay over time (Fig. 3b, top left), they did so at varying rates. Indeed, there is statistical support for strong collateral effects occurring at the beginning of the experiments for CTX and IMI (Pearson's $R = -0.68$ and -0.42 , respectively; $P < 0.01$ for both), but not for FOX ($R = -0.27$; $P = 0.068$). We then analysed collateral effects at the level of each individual mutational step, looking for correlations between MIC changes for CAZ versus the other antibiotics. Only for IMI did we observe a significant correspondence with CAZ ($R = -0.74$; $P < 10^{-7}$), whereas correspondence for CTX, and especially FOX, was virtually absent ($R = 0.21$ and 0.05 and $P = 0.180$ and 0.741 , respectively; Fig. 3b). The discrepancy between absolute, global effects and actual, individual

ones probably reflects the substantial degree of divergence across lines, with some developing cross-sensitivity and others developing cross-resistance (Fig. 3a).

Next, we assessed the extent to which epistasis shapes heterogeneity in collateral sensitivity, both among lines and over time within each line. Figure 3c shows that the endpoint sensitivity profiles varied markedly among groups, primarily due to H274 lines consistently ranking highest in MIC across antibiotics ($P < 0.002$; two-sample Wilcoxon test on ranked data). However, a few lines deviated from their group's typical profile (H3 and, to a lesser extent, S2). This deviation illustrates that, even when initial mutations markedly determine subsequent evolution, secondary mutations can cause pathways to branch away from the overall profile of the main route. Interestingly, in both H3 and S2, these deviations involved mutations that did not affect CAZ MICs (Supplementary Table 2), illustrating how cryptic changes during plateaus can unpredictably alter phenotypic profiles.

Next, we quantified how much divergence occurred over time along a single mutational path. For each line and secondary antibiotic combination, we classified first-step mutants as cross-resistant, neutral or cross-sensitive compared with the ancestor and then assessed how this classification changed for endpoint mutants. Figure 3d shows that the most common paths were those along which sensitivities changed classification (31%; 13/42) and those along which effects were aggravated (31%; 13/42), followed by paths with no change (21.4%; 9/42) and paths along which effects were mitigated (16.7%; 7/42). Notably, all paths with mitigating effects involved cross-sensitive first-step mutants.

Molecular traits rapidly diverge during adaptation

Thus far, we have interpreted our results in terms of MICs—readouts that reflect the aggregate contributions of multiple molecular-level traits. The traits typically considered to shape MICs the most are those directly affecting how efficiently an enzyme converts a substrate into a product (that is, the catalytic efficiency—the ratio of the turnover number over the substrate affinity (k_{cat}/K_m))^{49,78}. However, recent work revealed a major role for the multiple traits that affect the life cycle of β -lactamase, including its exit from the ribosome, translocation across the inner membrane, folding once in the periplasm and degradation by proteases^{79–83}. The intuition for this importance is that, regardless of how efficient the β -lactamase may be, these factors ultimately determine effective reaction rates by controlling the steady-state abundance of functional enzymes in their natural workplace⁸³.

Although it is difficult to pinpoint the specific molecular traits that are altered, an aggregate readout of all of these traits can be obtained by measuring abundance stability in the periplasm^{79,81}. Indeed, these measurements have successfully explained MIC differences among closely related enzyme variants⁸³ and, most revealingly, among the same enzyme across different hosts⁸². Moreover, beyond MICs, these traits influence fitness costs in the absence of drugs, as alterations in them can interfere with other cellular processes (for example, saturating the translation, membrane translocation and folding machinery) and trigger stress responses due to the aggregation of misfolded proteins^{84,85}. These costs in the absence of selective pressures for resistance are expected to be crucial for the long-term success of enzyme variants under ever-changing conditions^{82–84}.

To explore how epistasis may have driven divergence in molecular-level traits during CAZ adaptation, we measured catalytic efficiency (k_{cat}/K_m), periplasmic abundance stability and growth rates for some selected lines. First we examined catalytic efficiency in purified extracts (Methods), including all steps in the line with the largest MIC increase (H2), a deviating endpoint from this line (H1) and three high-MIC representatives from the other groups (D1, S1 and L2). The catalytic efficiency varied over three orders of magnitude (Fig. 4a), including one line with no increase relative to the ancestor (D179Y). These observations contrast with the observations that CAZ MICs

increased substantially in all lines and that endpoint MICs varied less than tenfold (512–4,096 $\mu\text{g ml}^{-1}$; Supplementary Table 2). Similar results were obtained using hydrolysis rate estimates from an alternative, spectrophotometric assay⁶⁰ (Extended Data Fig. 3 and Methods). Finally, we asked whether, at least at the level of each individual step, the effects on MIC and catalytic efficiency show any correlation (Fig. 4a, bottom). Consistent with previous studies^{49,81,83}, no significant correlation was detected ($R = 0.01$; $P = 0.98$).

Next, we examined the stability of enzyme abundance in the periplasm (that is, in-cell kinetic stability, as opposed to standard, in vitro thermal stability⁸⁶). To this end, we used western blots to quantify the change in the relative abundance of several evolved variants in the periplasm over time (Extended Data Figs. 4 and 5 and Methods). We chose two high-MIC lines from each group, including all of the individual mutants from each line. Periplasmic stability consistently dropped during the first steps of adaptation, later recovering to different extents in all lines, except for the D179-derived ones (Fig. 4b). In addition, although again the values for the endpoint isolates varied widely (four orders of magnitude), each individual mutation step showed a clear negative correlation between the effects on MIC and those on periplasmic stability ($R = -0.76$; $P < 10^{-6}$) (Fig. 4b, bottom). Interestingly, the largest individual effect on stability in this panel corresponds to a mutation showing no MIC change (L1, encoding the substitution E167G; Supplementary Table 2), reinforcing the idea that adaptive evolution continued after CAZ MICs plateaued. Interestingly, endpoint isolates from S171P-derived lines exhibited a markedly altered molecular weight at later incubation times (10 h) and this was also observed to a minor degree in other lines (Extended Data Fig. 4e). Proteomics confirmed the presence of the enzyme, suggesting potential structural or post-translational modifications that may enhance stability by facilitating binding to other components of the periplasm.

Although catalytic efficiency or periplasmic stability alone often fail to correlate with MIC increases, combining the two could result in a stronger positive correlation⁸³. Intuitively, for a given catalytic efficiency, higher stability should increase the MIC by ensuring greater enzyme availability. However, the combination of both traits still negatively correlates with MIC increase ($R = -0.91$; $P = 0.01$; Extended Data Fig. 6). One explanation is that expected synergy only holds if no direct trade-offs exist between these traits (as in ref. 83, which compares different β -lactamases). Yet, gain-of-function mutations leading to an enhancement of catalytic activity often simultaneously reduce stability⁶⁰—a trade-off that was evident among our first-round mutants (Fig. 4b). Such trade-offs typically arise from suboptimal rearrangements following active-site remodelling and the increased structural strain caused by the higher catalytic turnover⁶⁰. Moreover, the magnitude of stability loss in our system greatly exceeds efficiency gains, leading the combined metric to correlate negatively with MIC (Extended Data Fig. 6).

We then measured growth rates as a proxy for collateral fitness costs resulting from the evolved CAZ resistance^{84,85}. We conducted the assays at subinhibitory concentrations of CAZ, not only to be able to include the ancestor but also to assess costs under conditions where enzymes are still required to perform their physiological function⁸⁷. First we estimated growth rates for the same two high-MIC lines per group used in the periplasmic stability study, including all intermediate mutants. Growth rates improved over rounds (Fig. 4c and Extended Data Fig. 7), as expected due to the higher protective capacity of the evolved enzymes. However, endpoint growth rates exhibited variations that did not align with MIC differences. For instance, line L1 displayed the highest growth rate despite having the lowest MIC (256 $\mu\text{g ml}^{-1}$; Supplementary Table 2). Similarly, S171-derived lines showed the lowest growth rates despite displaying some of the highest MICs (1,024 and 2,048 $\mu\text{g ml}^{-1}$ for S1 and S2, respectively; Supplementary Table 2). At the level of each individual mutation step, MIC and growth effects correlated positively ($R = 0.79$; $P < 10^{-11}$; Fig. 4c, bottom). Of note, line

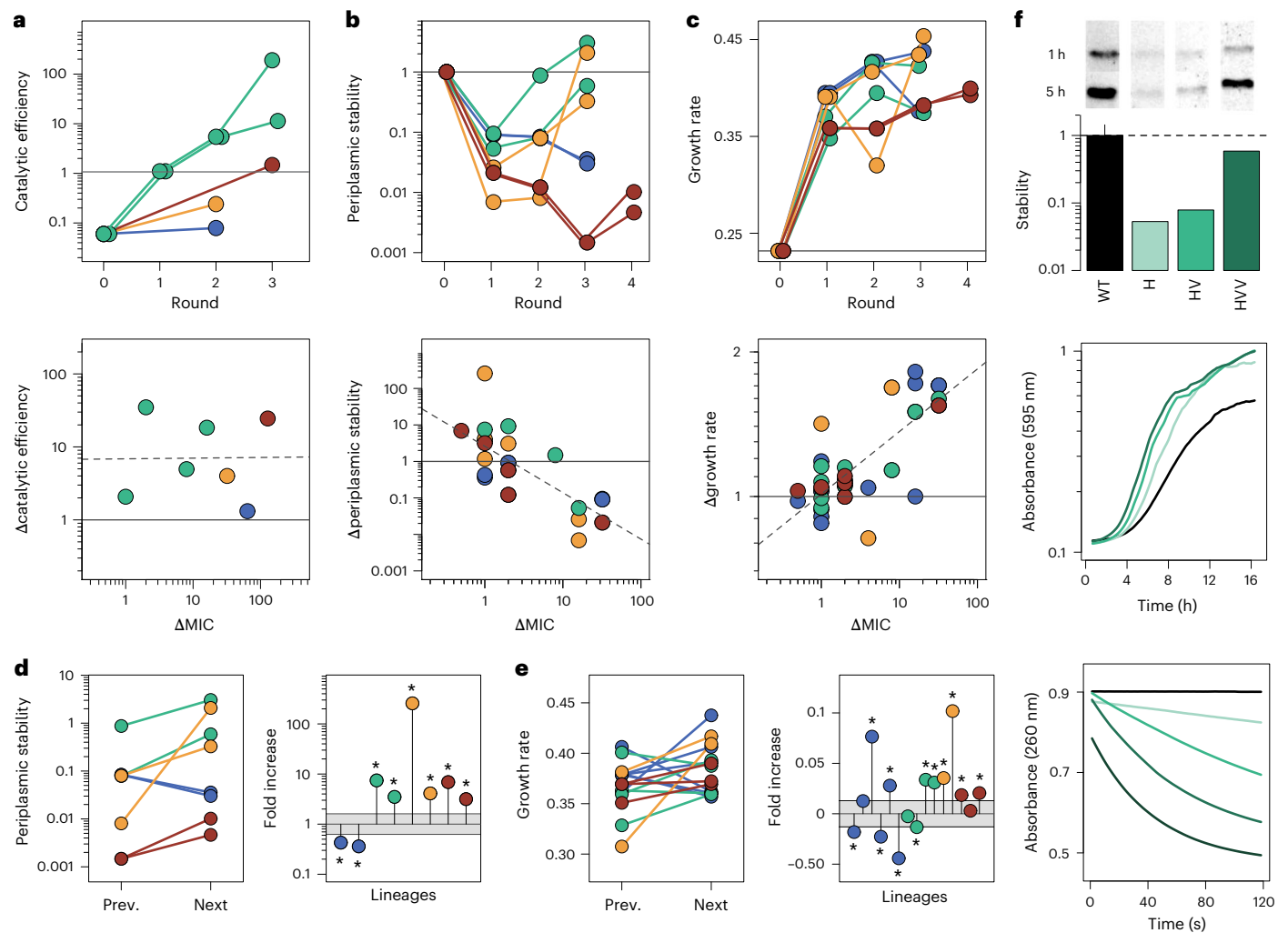


Fig. 4 | Molecular traits rapidly diverge during CAZ resistance evolution. **a**, Top, catalytic efficiency (k_{cat}/K_m) across rounds for selected enzyme variants, expressed as a fold change relative to the ancestral KPC-2 enzyme. The colours indicate initial CAZ resistance variants: blue, green, yellow and red for D179Y, H274Y, L169P and S171P, respectively. Bottom, correlation between individual changes in k_{cat}/K_m and MIC, with the best fit represented by a dashed line and the ancestral value represented by a solid line. Throughout the panels, the delta symbol indicates step-wise changes relative to the immediate ancestor. For non-green points in this panel, catalytic efficiency data from the immediate ancestor were unavailable; thus, delta values were calculated relative to ancestral KPC-2. **b**, Top, stability of enzyme abundance in the periplasm across rounds for the same variants as in Fig. 3. The values represent western blot intensity changes over 5 h of growth, expressed relative to the ancestral KPC-2 enzyme (see f). The colour coding is as for a. Bottom, correlation between individual changes in periplasmic stability and MIC. The dashed line indicates the best fit from linear regression and the horizontal solid line represents the stability of the ancestral enzyme. **c**, Top, growth rates in sub-MIC concentrations of CAZ across rounds

for the same variants as in b. Bottom, correlation between individual changes in growth rate and MIC. The dashed and solid lines show best fit and reference values, respectively, as above. **d**, Left, change in periplasmic stability between consecutive mutants within plateaus. Right, change expressed as a fold increase. The grey bar represents the control's 95% confidence interval (mean $\pm$ 2 s.e.m.). Asterisks mark values outside of this range ($P < 0.05$; $n = 6$). **e**, As in d, but showing changes in the growth rate between consecutive mutants within each plateau ($n = 5$). **f**, Top, example western blot quantifying periplasmic stability, with a bar plot showing the log intensity ratio of initial (1 h) and final (5 h) bands for each variant. Labels refer to the first letter of the amino acid substitutions acquired at each round along line H1 (H274Y, V240A and V103L (Supplementary Table 1). The error bar represents the control's 95% confidence interval across gels (mean $\pm$ 2 s.e.m.; $n = 8$; Extended Data Fig. 4a–d; note that all others are single measurements). Middle, example growth rates in sub-MIC CAZ concentrations for the same variants. Bottom, example CAZ concentration decay from the colorimetric assay for the same variants, plus the closely related variant from round 3, H2 (darkest green; Supplementary Table 1). WT, wild type.

L2 underwent a large growth rate decrease in round three, followed by full recovery in the next round. This recovery involved a mutation not altering the MIC (encoding the substitution E168D; Supplementary Table 2), again pointing at the non-static nature of plateaus.

We also measured growth rates under CAZ-free conditions, as substantial differences in the absence of drug may greatly influence the long-term spread of resistance enzymes across hosts and environments. Extended Data Fig. 8 reveals clear fitness differences among strains, with some first-round mutants growing markedly slower. Notably, these costs were largely alleviated over time, with some plateau

isolates becoming even fitter than the ancestral strain—a worrisome perspective from a clinical standpoint. Overall, these differences suggest that divergence among mutants affects multiple molecular traits, potentially involving energetic inefficiencies, competition for transcriptional and translational resources or interference with other proteins or macromolecules^{84,85}.

Finally, we examined changes occurring specifically during the plateaus. In all lines except those derived from D179, the periplasmic stability increased between consecutive mutants within each plateau (6/8; Wilcoxon signed-rank test; $P = 0.02$; Fig. 4d). The growth rate

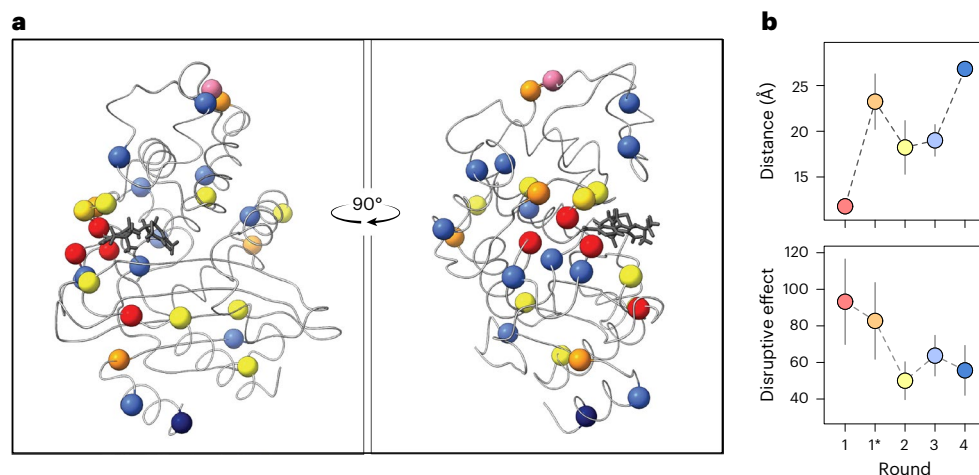


Fig. 5 | Positions and properties of accumulated mutations across rounds. **a**, Schematic of the KPC-2 β -lactamase backbone with CAZ bound at the active site (shown as sticks). Positions where alterations were observed following mutation are represented as spheres, coloured according to the round in which they arose (red for those encoded by presumed driver mutations in round 1, orange for those encoded by accessory mutations in round 1* in **b**), yellow for round 2, light blue for round 3 and dark blue for round 4). The purple

sphere marks the P94 substitution, present in all rounds of mutagenesis. **b**, Top, average atomic distances of amino acid alterations from CAZ's centre of mass. Bottom, Grantham scores (physiochemical distances) as a proxy of the disruptive effect of each alteration per round. The circle colours are as in **a**. The error bars represent standard deviation (top, left to right: $n = 5, 6, 9, 13$ and 1; bottom: $n = 5, 7, 10, 14$ and 3; note that atomic distance is not defined for substitutions in the peptide signal). Panel **a** generated using ChimeraX.

also increased in most lines (8/15), a marginally significant trend that became significant when excluding D179-derived lines (Wilcoxon signed-rank test; $P = 0.054$ and 0.0136 , respectively; Fig. 4d and Extended Data Figs. 4 and 5). Overall, these results suggest that adaptive evolution continues across multiple molecular traits even after CAZ MICs stabilize. Moreover, the locations and predicted effects of mutations over rounds provide further support for this notion: initial mutations encode disruptive substitutions near the active site, probably reshaping its physical and chemical properties to enhance enzymatic activity (Fig. 5). Later mutations tend to encode substitutions that occur further away and have milder effects, possibly compensating for the stability loss and collateral fitness costs incurred by the initial mutations. Similar patterns of mutations encoding amino acid changes falling progressively further away from the active centre have been observed in other proteins^{88,89}.

Discussion

In this study, we assessed the extent to which strong epistasis can drive divergence not only at the genotypic level, but also across phenotypes that may be relevant under alternative conditions. We first characterized the diversity of mutational paths through which KPC-2 carbapenemase can evolve increased activity towards a poor substrate, the β -lactam antibiotic CAZ. Our results show that KPC-2 can readily adapt through different routes with markedly divergent phenotypes, including differences in catalytic efficiency, periplasmic stability, collateral fitness costs and susceptibility to other antibiotics. The speed and degree of divergence is remarkable considering that we conducted the experiments under identical laboratory conditions, and bears significant implications for designing strategies to prevent antibiotic resistance, as well as for understanding natural evolution in complex environments.

Why do the initial CAZ resistance mutations lead to such divergent outcomes? We classified the lines into four groups based on the positions of the amino acid substitutions encoded by their first-step mutations. These positions varied in how frequently they were altered among natural variants: position H274 retrieved the most hits (47%) in the Beta-Lactamase DataBase (<http://bldb.eu>)⁵⁹, followed by D179 (26%), L169 (15%) and S171 (9%). Reflecting this unequal prevalence, detailed mechanistic studies are available only for positions D179 and

H274. Substitutions with N and Y at D179 increase Ω -loop flexibility, facilitating the rapid entry of CAZ into the active site^{90,91}. However, this flexibility decreases the CAZ hydrolysis rate due to the displacement of essential hydrolytic residues. This is consistent with the catalytic efficiency reductions we observed (Fig. 4a,b and Extended Data Fig. 3). Interestingly, resistance in D179 variants is attributed to a kinetic trap, whereby the swift entry of CAZ into the reaction pocket reduces to safe levels the amount of free molecules available for inhibiting penicillin-binding proteins⁹¹. In contrast, substitution with Y at H274 enhances hydrogen bonding with CAZ's oxymino side chain, directly increasing the hydrolysis rate⁶⁰. Additionally, H274Y facilitates CAZ entry into the active site without altering the Ω loop, by repositioning the surrounding 240 and 270 loops⁹². To sum up, these cases represent fundamentally different mechanisms for enhancing β -lactamase activity, making it reasonable to expect that they can produce distinct phenotypic profiles.

One key implication of our findings pertains to current efforts to exploit collateral sensitivities to prevent antibiotic resistance⁹³. We show that widespread strong epistasis can drive rapid, substantial divergence of resistance profiles across different antibiotics, even in parallel replicates evolving under the same conditions. Moreover, this divergence can occur not only among lines initiated from different mutations, but also along the sequential steps that comprise each line. These observations reveal extra challenges in anticipating evolutionary responses to antibiotics^{28–33,35}. For instance, resistance trade-offs may markedly differ among seemingly similar populations, due to them being dominated by the descendants of different mutants that happened to be the first to occur. Furthermore, heterogeneities in resistance are also expected among individuals within a single population, provided its size is sufficiently large. To further complicate matters, trade-offs observed early may not persist through further adaptation. Together, these results challenge the reliability of snapshot characterizations of typical mutants as indicators of populations' responses to future treatments²⁸. Thus, this work adds to the cautionary evidence suggesting that collateral sensitivity strategies, although promising, need to be carefully tailored to consider the many factors influencing the spectrum of adaptive outcomes^{28–33,35}.

Our findings also bear implications for understanding the natural evolution of antibiotic resistance enzymes, including their

geographical and phylogenetic distributions. Clinical pathogens often face multiple antibiotics with concentrations that change over space and time^{87,94–96}. Since phenotypic profiles are strongly determined by the chance occurrence of initial mutations, which variants will prevail in a particular setting will be heavily contingent on the specific order and intensity of antibiotic exposures. Small population sizes or transmission bottlenecks will further amplify this historical contingency, simply due to the stochastic sampling of variants with distinct properties in other environments^{28,50}.

Beyond differences in antibiotic resistance profiles, we have shown that strong epistasis can readily drive divergences in periplasmic abundance and collateral fitness costs—traits that presumably reflect how enzymes interact with various cellular components throughout their life cycle^{82–85}. These differences are probably critical for the long-term success of variants during their journey across multiple hosts^{82,83,97}. For one thing, selection will favour the least costly variants under low or no antibiotic pressure^{84,85}. Moreover, although our evidence comes from a single host, these differences may be amplified or mitigated in other hosts with their own idiosyncrasies in their cellular components. Thus, strong epistasis might drive the rapid association of certain variants with specific hosts, an intriguing possibility given the major role of horizontal gene transfer in the dissemination of resistance genes^{44,45}.

Lastly, it is pertinent to ask whether the strong epistasis-driven phenotypic divergence observed here with KPC-2 is specific to this enzyme, or if similar patterns can be expected in other systems. KPC-2 has been hypothesized to be more evolvable than other class A β -lactamases due to its notably higher thermal stability (66.5 °C for KPC-2 versus 51.5 °C for TEM-1 and 51 °C for CTX-M)⁶⁰. High thermal stability enables the accumulation of destabilizing beneficial mutations without compromising folding and functionality, a property shown to boost evolvability in different enzymes^{98,99}. Thus, compared with other β -lactamases, KPC-2's superior stability may facilitate the acquisition of first-step mutations with a more diverse range of molecular mechanisms for increasing resistance, being thus more prone to rapid divergence across multiple traits.

Beyond single proteins, it remains to be determined whether epistasis-driven phenotypic divergence is a general feature of rapid bacterial adaptation. For instance, in a long-term evolution experiment with *E. coli*, divergence across secondary environments was moderate and took thousands of generations to manifest^{100,101,102}. This was perhaps because selection pressures in long-term evolution experiments are milder and adaptation involves many genes, so the outcomes are not dominated by the biophysical details of a single enzyme. Supporting this interpretation, a similar experiment that adapted 114 lines of *E. coli* to a nearly lethal temperature found much greater divergence within just 2,000 generations¹⁰². Notably, some lineages led to resistance against rifampicin, an antibiotic never present in the medium¹⁰³, a clear illustration of consequential divergence during genome-wide evolution. Further research is required to clarify the determinants of epistasis-driven divergence across environments.

Methods

Bacterial strains and plasmids

The *E. coli* TOP10 strain (Invitrogen) and the pUC-derived, high-copy-number pCR-Blunt II-TOPO vector (Invitrogen), carrying a kanamycin resistance gene, were used for the random mutagenesis experiments, susceptibility testing, growth curves and periplasm extract collection. The *E. coli* Origami 2(DE3) strain (Novagen) and pET-28a plasmid (Novagen) were used for protein expression and purification.

Random mutagenesis

The wild-type *bla*_{KPC-2} gene was amplified from a reference strain obtained from the laboratory of T. Naas¹⁰⁴ (env-KPC-2_Fw (5'-TTCAAACAAGGAATATCGTTGATGTCAGTATCGCCGTCT-3') and env-KPC-2_Rv (5'-AATAGATGATTTTCAGAGCCTTACTGCCCGTTGACGCCCA-3')),

cloned into the pCR-Blunt II-TOPO vector (Invitrogen) and expressed in *E. coli* TOP10 (Invitrogen). Random mutagenesis was performed using the GeneMorph II EZClone Domain Mutagenesis Kit (Agilent) according to the manufacturer's instructions. The *bla*_{KPC-2} gene was amplified with a similar pair of primers as above, but sparing the start of the open reading frame (PRE-KPC-2_Fw (5'-TTCAAACAAGGAATATCGTTG-3') and PRE-KPC-2_Rv (5'-AATAGATGATTTTCAGAGCC-3')) and using Mutazyme II DNA polymerase, which according to the manufacturer generates a low-bias mutation spectrum with similar mutation rates at As/Ts and Gs/Cs. To achieve a low mutation rate (0–4.5 mutations per kilobase), 500 ng target DNA and 30 PCR cycles were used. The resulting variant library was purified using the GeneJET PCR Purification Kit (Thermo Fisher Scientific). A second PCR, using the mutant library as megaprimers and the pCR-Blunt II-TOPO (*bla*_{KPC-2}) plasmid as the template, was performed. In the next step, the template was digested with DpnI (Thermo Fisher Scientific) and the PCR product was transformed into *E. coli* TOP10. Transformants were incubated overnight at 37 °C in LB broth supplemented with 50 $\mu\text{g ml}^{-1}$ kanamycin to ensure plasmid retention. Mutant selection was performed in Mueller Hinton broth (Thermo Fisher Scientific). Four tubes containing 20 ml Mueller Hinton broth with increasing concentrations of CAZ (Thermo Fisher Scientific) were used. The first concentration corresponded to the midpoint of the MIC conferred by KPC-2, and subsequent concentrations were doubled. Inocula were adjusted to 5×10^5 c.f.u. ml^{-1} and cultures were incubated overnight at 37 °C. The tube with the highest CAZ concentration showing visible growth was plated on LB agar with 50 $\mu\text{g ml}^{-1}$ kanamycin. For the first mutagenesis round, two randomly selected colonies were sent for Sanger sequencing using M13 primers (M13_Fw (5'-GTAAAACGACGGCCAG-3') and M13_Rv (5'-CAGGAACAGCTATGAC-3')), which were also used in subsequent rounds.

Susceptibility testing

MIC values for *E. coli* TOP10 strains carrying KPC variants were determined using the broth microdilution method following European Committee on Antimicrobial Susceptibility Testing guidelines. In brief, 96-well plates were used with a final volume of 200 μl per well. In the first well, 100 μl Mueller Hinton broth and antibiotic stock (to a final concentration of <25% of 100 μl) were added to achieve the highest concentration tested. Serial twofold dilutions of the antibiotic were then prepared. Finally, 100 μl inoculum (1×10^6 c.f.u. ml^{-1}) from an overnight culture was added. MIC values were assessed after 18 h of incubation at 37 °C.

Site-directed mutagenesis

To generate all combinations of four key point mutations, we designed standard primers for the V240A, L169P, H274Y and D179Y substitutions using the QuikChange Primer Design tool (Agilent). The *bla*_{KPC-2} gene cloned into the pCR-Blunt II-TOPO vector (Invitrogen) was used as the template for mutagenesis, which was performed with the QuikChange II Site-Directed Mutagenesis Kit (Agilent) following the manufacturer's instructions. Mutagenesis products were transformed into *E. coli* TOP10, and transformants were selected on LB agar with 50 $\mu\text{g ml}^{-1}$ kanamycin. Plasmids containing single-point KPC-2 variants were then used as templates for subsequent rounds of mutagenesis to sequentially introduce additional mutations, ultimately generating all 16 combinations. Each construct was verified by Sanger sequencing.

Growth curves

A 300- μl volume of LB broth containing either 8 $\mu\text{g ml}^{-1}$ CAZ or 50 $\mu\text{g ml}^{-1}$ kanamycin was inoculated with *E. coli* TOP10 expressing KPC-2 and its variants and placed in 100-well honeycomb plates. Incubation was carried out at 37 °C for 20 h using the Bioscreen C Pro Microbiological Growth Analyser (LabSystems), with optical density at 600 nm (OD_{600}) values measured every 10 min. Five replicates were used per variant. Growth rate values were determined from the maximum slope of the natural logarithm of optical densities over time.

Protein purification

The *bla*_{KPC-2} gene and variants' fragments corresponding to the mature β -lactamase were cloned into the expression vector pET-28a, using the PCR-generated fragment with the primers NdeI-KPC-2-sps_Fw (5'-CATATGCTGACCAACCTCGTCGCGGA-3') and BlnI-KPC-2_Rv (5'-GCTCAGCTTACTGCCCGTTGACGCCCAAT-3') and the NEBuilder HiFi DNA Assembly Cloning Kit (New England Biolabs), following the manufacturer's instructions. Recombinant plasmids were transformed into chemocompetent *E. coli* Origami 2(DE3) and the transformants were selected in LB agar plates supplemented with 50 μ g ml⁻¹ kanamycin. An overnight culture of *E. coli* Origami 2(DE3) harbouring pET-28a(*bla*_{KPC-2} or mutated *bla*_{KPC}) was used to inoculate 2 l LB broth containing 50 μ g ml⁻¹ kanamycin. Bacteria were cultured at 37 °C until an OD₆₀₀ of 0.6 was reached. Expression of KPC-2 and its variants was induced overnight at 25 °C with 0.1 mM isopropyl β -D-1-thiogalactopyranoside. The *bla*_{KPC-2} gene and its mutated versions were purified in one-step pseudo-affinity chromatography using an NTA-Nickel column (Cytiva). Protein purity was estimated by sodium dodecyl sulfate polyacrylamide gel electrophoresis (SDS-PAGE) and pure fractions were pooled and dialysed against 20 mM HEPES SO₄K₂ 50 mM buffer (pH = 7) and concentrated using Vivaspinn columns (Cytiva). Protein concentrations were determined by Bradford protein assay (Bio-Rad).

Steady-state parameters

The kinetic parameters of purified KPC-2 and its variants were determined at 25 °C in 100 mM sodium phosphate buffer (pH = 7). The k_{cat} and K_m values were determined by analysing the hydrolysis of β -lactams under initial-rate conditions with an Ultrospec 2000 model UV spectrophotometer (GE Healthcare). Kinetic parameters were calculated using the nonlinear regression of the Michaelis–Menten equation $v = k_{cat}[S]/(K_m + [S])$, where v is the initial reaction velocity and $[S]$ is the substrate concentration. The β -lactams were purchased from Sigma–Aldrich. The velocity of CAZ hydrolysis could not be saturated by measurable concentrations due to a high K_m . Thus, the second-order rate constant at steady state, k_{cat}/K_m , was determined by fitting the progress curves to the equation $v = k_{cat}/K_m[E][S]$, where $[E]$ is the enzyme concentration and $[S] \ll K_m$ (ref. 60).

Periplasmic stability

A C-terminal haemagglutinin tag was added to the proteins expressed by *E. coli* TOP10 strains harbouring pCR-Blunt II-TOPO (*bla*_{KPC-2} gene and variants) using PRE-KPC-2_Fw and KPC-HA-STOP_Rv primers (5'-TTCAAACAAGGAATATCGTTG-3' and 5'-TTATGCATAATCCGGAACATC ATACGGATACTGCCCGTTGACGCC-3', respectively). Extraction of periplasmic proteins was adapted from ref. 86. *E. coli* TOP10 pCR-Blunt II-TOPO strains (*bla*_{KPC-2} gene and variants) were incubated in LB broth supplemented with 50 μ g ml⁻¹ kanamycin until an OD₆₀₀ value of 0.30 was reached, which was taken as t_0 . Aliquots of the cultures (2 ml) were then taken at 1, 2, 5 and 10 h from t_0 . The aliquots were pelleted and washed with 20 mM Tris-HCl (pH 8.5) and 150 mM NaCl. To normalize the number of cells, the volume according to $V = (2 \text{ ml} \times 0.064 \text{ OD}_{600})/\text{OD}_{600}$ was resuspended in 1 ml 20 mM Tris-HCl (pH 8.5), 0.1 mM EDTA, 20% wt/vol sucrose, 1 mg ml⁻¹ lysozyme (Sigma–Aldrich; protein $\geq 90\%$) and 0.5 mM phenylmethylsulfonyl fluoride and incubated with gentle agitation at 4 °C for 30 min. After centrifugation at 15,000g for 2 min at 4 °C, the supernatant containing the periplasmic fraction was collected and stored at 4 °C. Levels of C-terminal haemagglutinin (HA)-tagged proteins were determined in the periplasmic fraction by SDS-PAGE and transferred to iBlot 3 Transfer Stacks, nitrocellulose (Thermo Fisher Scientific), followed by western blot with rat anti-HA high affinity monoclonal antibodies (Sigma–Aldrich) at 1:1,000 dilution for KPC-2 and its variants. Goat anti-rat immunoglobulin horseradish peroxidase (Invitrogen; 1:10,000 dilution) was used as a secondary antibody. SuperSignal West Pico PLUS Chemoluminescent Substrate

(Thermo Fisher Scientific) was used for detection. Protein band intensities were quantified with ImageJ. The Precision Plus Protein Standards Kaleidoscope (Bio-Rad) provided molecular weight standards.

Proteomics analysis

Protein identification and characterization by liquid chromatography with tandem mass spectrometry (MS/MS) was carried out at the Centro de Biología Molecular Severo Ochoa (Consejo Superior de Investigaciones Científicas) protein chemistry facility (Madrid, Spain). Briefly, in-gel digestion was performed after SDS-PAGE of the periplasmic extract. Gel bands of the target molecular weight were dried, destained in acetonitrile (1:1), reduced with 10 mM dithiothreitol at 56 °C for 1 h and then alkylated with 10 mM iodoacetamide for 30 min at room temperature in darkness. The proteins were then digested in situ with sequencing-grade trypsin (Promega), following the method described by Pérez et al.¹⁰⁵. The gel pieces were dehydrated with acetonitrile, then dried in a speedvac. They were rehydrated in 100 mM Tris-HCl (pH 8) and 10 mM CaCl₂ with 12.5 ng μ l⁻¹ trypsin for 1 h on ice. After removing the digestion buffer, the gels were incubated with fresh 100 mM Tris-HCl (pH 8) and 10 mM CaCl₂ at 37 °C for 12 h. Digestion was stopped by adding 1% trifluoroacetic acid. The supernatants were dried and desalted using ZipTip C18 pipette tips (Millipore) before mass spectrometry analysis.

The desalted protein digest was resuspended in 10 μ l 0.1% formic acid and analysed by reverse-phase liquid chromatography coupled with MS/MS using an Easy-nLC II system and an LTQ Orbitrap Velos Pro mass spectrometer (Thermo Fisher Scientific). Peptides were concentrated on-line with a 0.1 mm $\times$ 20 mm C18 RP precolumn (Thermo Fisher Scientific) and separated using a 0.075 mm $\times$ 250 mm bioZen 2.6 μ m Peptide XB-C18 RP column (Phenomenex) at 0.25 μ l min⁻¹. A 60-min dual gradient was applied: 5–25% solvent B for 45 min, 25–40% solvent B for 45 min, 40–100% solvent B for 2 min and 100% solvent B for 18 min (solvent A: 0.1% formic acid in water; solvent B: 0.1% formic acid and 80% acetonitrile in water). Electrospray ionization was performed with a nano-bore stainless steel emitter (Proxeon) at 2.1 kV, with the S-lens set at 60%. The Orbitrap resolution was 30,000. Survey scans (400–1,600 atomic mass units) were followed by 20 data-dependent MS/MS scans (top 20), with an isolation width of 2 atomic mass units, a normalized collision energy of 35% and dynamic exclusion for 60 s. Charge-state screening rejected unassigned and singly charged ions.

Peptide identification from raw data was carried out using the PEAKS Studio XPro search engine (version 11.5; Bioinformatics Solutions). A database search was performed against the UniProt reference proteome for *Escherichia coli* K-12 strain MG1655 (UP000000625), supplemented with the KPC-2-HA sequence, using a standard target-decoy fusion approach. The following constraints were used for the searches: a tryptic cleavage after Arg and Lys; up to two missed cleavage sites; and tolerances of 20 ppm for precursor ions and 0.6 Da for MS/MS fragment ions. The searches were performed allowing optional Met oxidation and Cys carbamidomethylation. False discovery rates for peptide spectrum matches and protein were limited to 0.01. Only proteins with at least two unique peptides discovered from liquid chromatography coupled with MS/MS analyses were considered to have been reliably identified.

Statistical analyses and data visualization

All of the statistical analyses were conducted using the R programming language (version 3.6.3), utilizing built-in functions and available packages. Pearson correlation coefficients, linear regression, Student's *t*-tests, Wilcoxon rank-sum tests and pairwise Wilcoxon signed-rank tests were performed with the functions `cor()`, `lm()`, `t.test()`, `wilcox.test()` and `pairwise.wilcox.test()`, respectively. For all Pearson correlation analyses, we tested the normality of residuals using the one-sample Kolmogorov–Smirnov test, as implemented by the function `ks.test()`.

In all cases, the results did not deviate from the null hypothesis of normality ($P > 0.073$). Data visualization for Figs. 1c and 3c used the *rgl* and *pheatmap* libraries, respectively.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All of the data needed to replicate the findings of this study are available in the manuscript and its Supplementary Information. Source data are provided with this paper.

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Author contributions

L.D. and A.C. conceived of the study idea. L.D. designed the methodology. L.D. performed the investigation with support from I.N. A.C. performed the formal analysis, supervised the study, wrote the original draft of the manuscript and acquired funding with support from L.D. A.C. and L.D. contributed equally to reviewing and editing the manuscript, with support from I.N.

Competing interests

The authors declare no competing interests.

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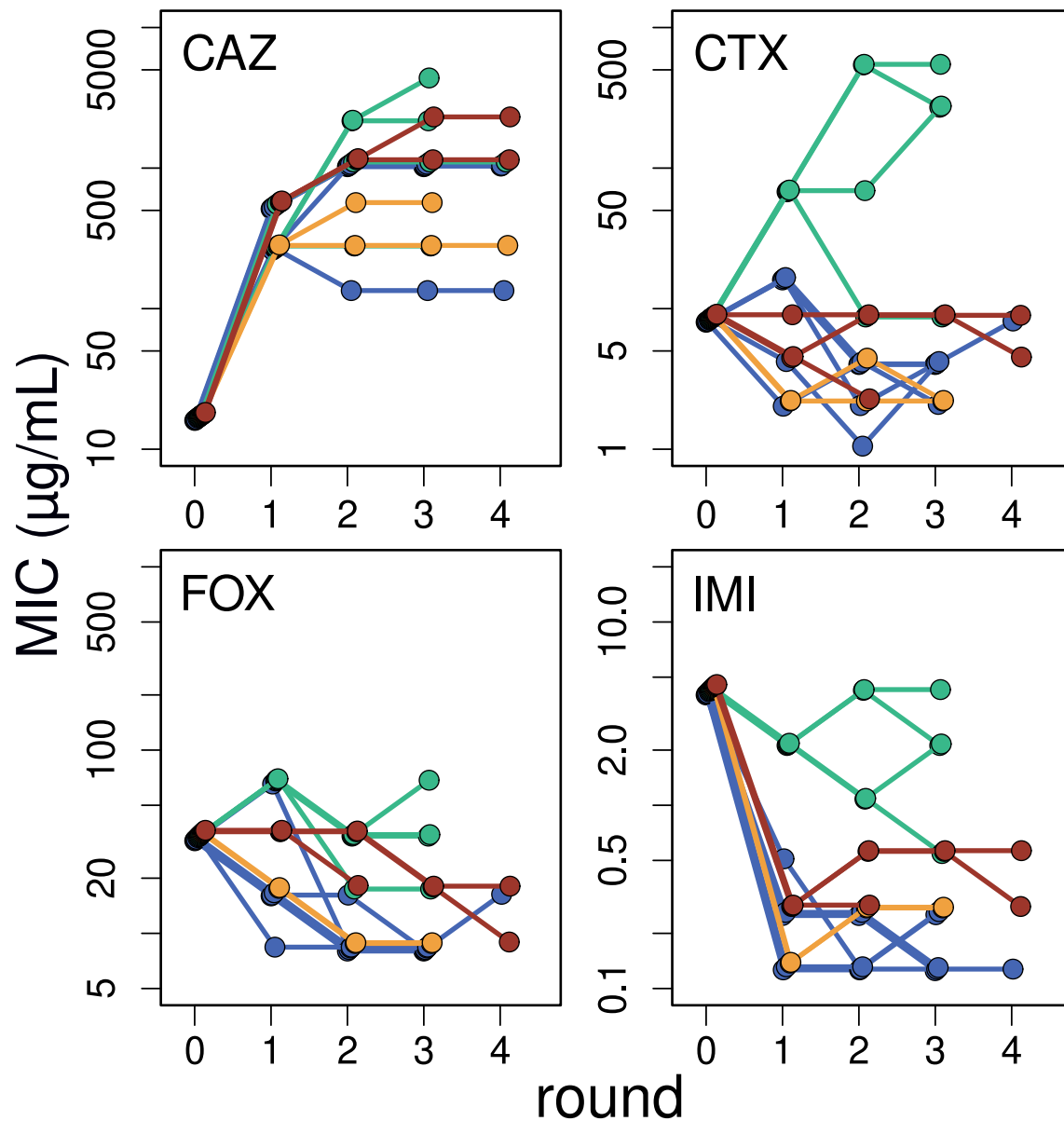
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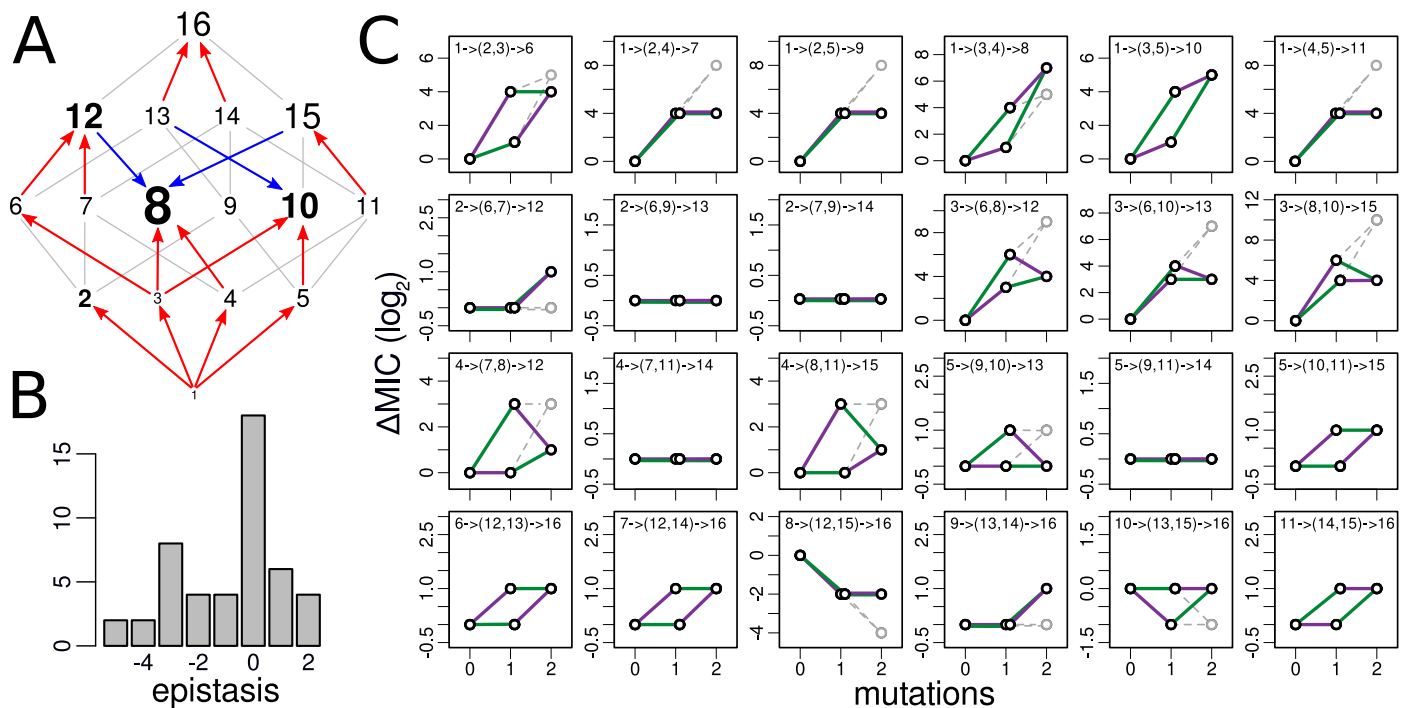
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Extended Data Fig. 1 | Resistance trajectories plotted against the number of rounds for each lineage. Colors indicate grouping based on positions associated with CAZ resistance: blue (D179Y), green (H274Y), yellow (L169P), and red (S171P). Panels show MIC values for CAZ, cefotaxime (CTX), ceftazidime (FOX), and

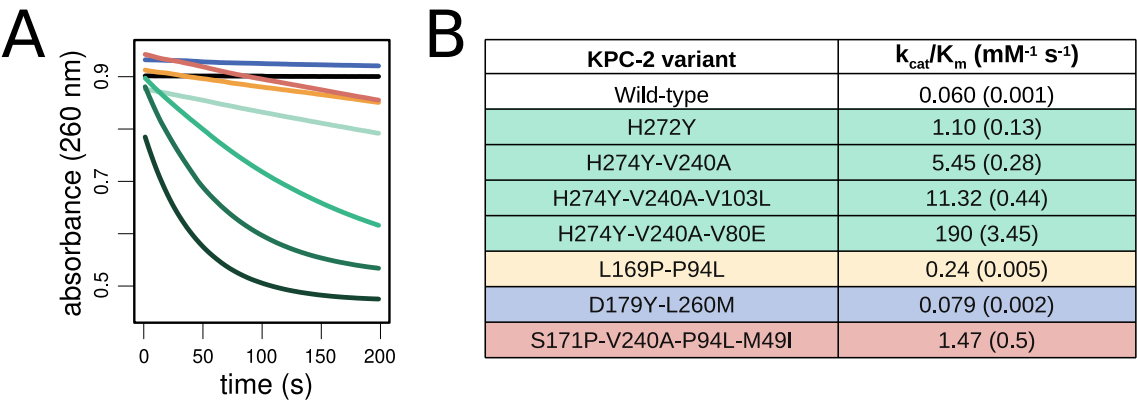
imipenem (IMI), as labeled in the upper-right corner. Note that divergence in resistance profiles peaks around rounds 1–2, similarly to other traits (Fig. 3), underscoring how rapidly populations diverge after fixation of the first mutation.



Extended Data Fig. 2 | Pairwise epistasis across the four-mutation landscape.

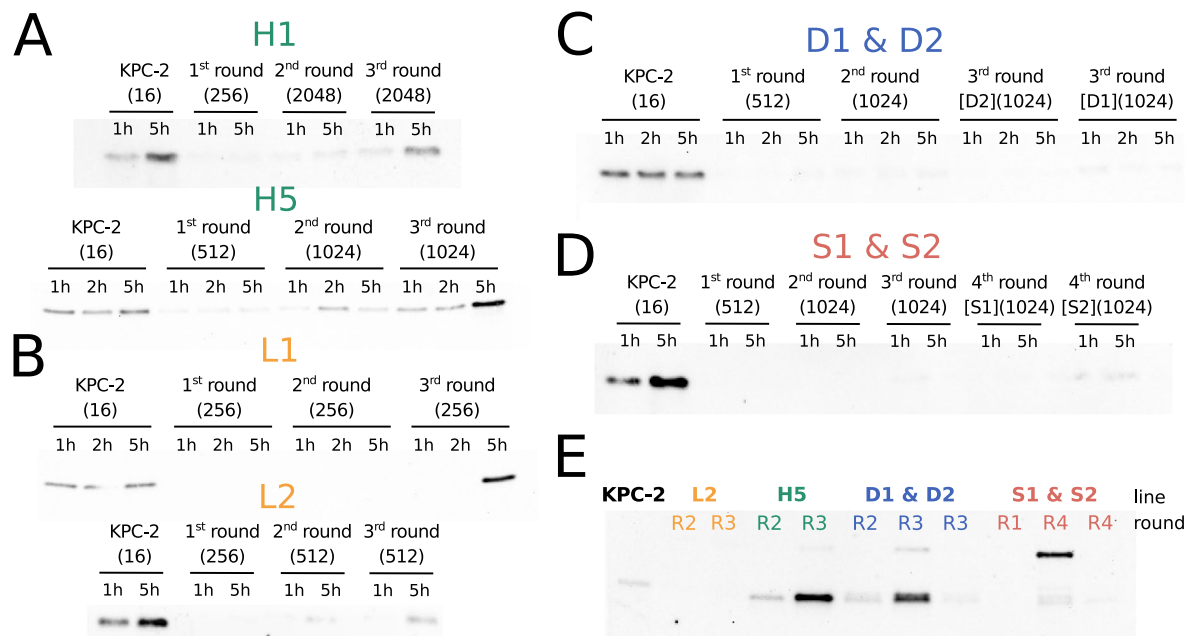
(a) Diagram of all mutational trajectories from the ancestral KPC-2 (bottom node). Conventions follow those in Fig. 2a, except that genotypes are numbered 1–16 (rather than labelled by the first letters of mutations) to facilitate the identification of genotypes shown in panel C. Genotype positions match those shown in Fig. 2a. (b) Distribution of epistasis values, calculated as the

deviation of each double-mutant fitness from the additive expectation. (c) All possible pairwise interactions. At the top of each panel it is indicated which four genotypes are being considered. For example, the top-left panel compares the single effects of L169P and V240A on the ancestor (genotypes 1, 2 and 3) with their combined effect (genotype 6).



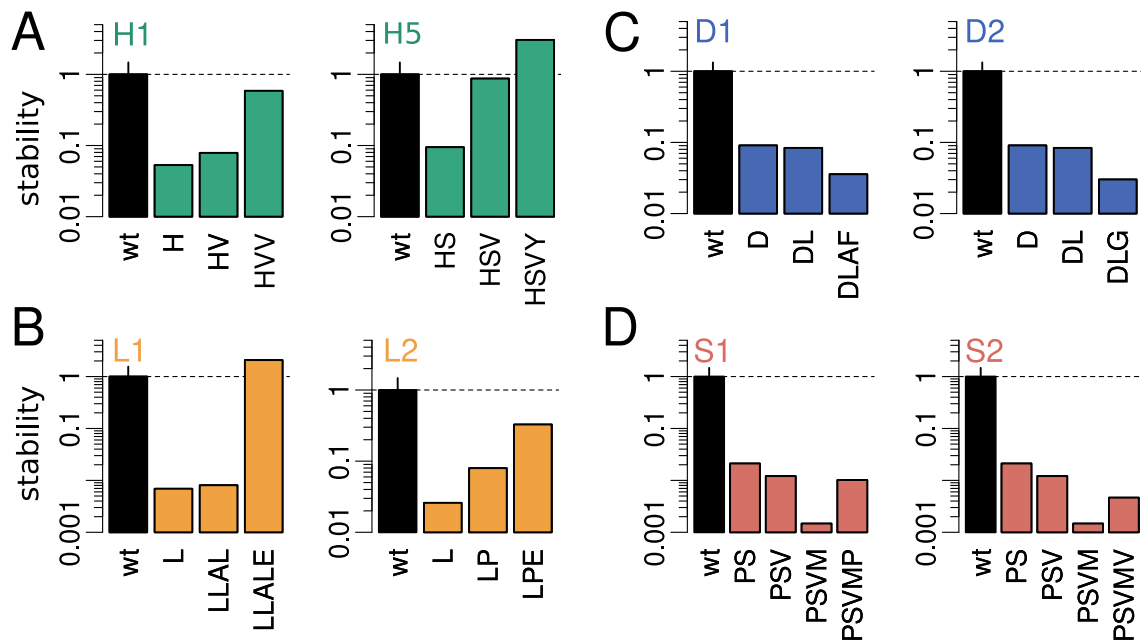
Extended Data Fig. 3 | Biochemical characterization of selected enzymes. (a) Progress curves of CAZ hydrolysis from a spectrophotometric assay. All reactions were performed with 500 nM enzyme and 50 μM ceftazidime. Hydrolysis of ceftazidime results in a decrease in absorbance at 260 nm. Curves correspond to the following variants: wild-type (black), D179P-L260M (blue),

L169P-P94L (yellow), S171P-V240A-P94L-M49I (orange), H274Y (very light green), H274Y-V240A (light green), H274Y-V240A-V103L (green), and H274Y-V240A-V80E (dark green). (b) Steady-state kinetic parameters for CAZ hydrolysis for ancestral KPC-2 and the same variants as in A. Standard errors are provided in parentheses ($n = 3$).



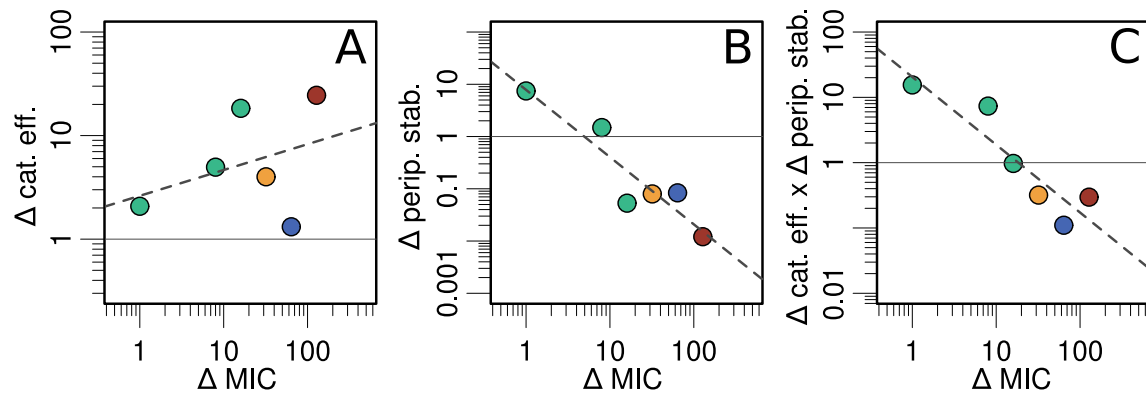
Extended Data Fig. 4 | Western blot of KPC-2 and its variants in periplasmic extracts of *E. coli* TOP10. (a–d) Periplasmic abundance of representative lineages from groups H274 (a, green), L169 (b, yellow), D179 (c, blue), and S171 (d, red) after 1 and 5 h of growth. (e) Immunoblot of selected KPC variants in the periplasm of *E. coli* TOP10 after 10 h of incubation. Isolates from each lineage are identified by the top label and color-coded as in previous panels. The bottom

label indicates the evolutionary round of each isolate. Note that for groups D1 & D2 and S1 & S2, the earliest isolate shown (R2 and R1, respectively) is ancestral to both descendant lineages. Repeated round labels (R3 and R4, respectively) represent isolates sampled after lineage splitting and are arranged in natural alphanumeric order (for example, round 3 D1, round 3 D2).



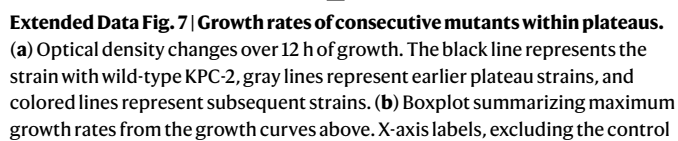
Extended Data Fig. 5 | Periplasmic stability of selected variants. Bar plots display the log intensity ratio of initial (1 h) and final (5 h) bands for each variant, quantified from the Western blots in Extended Data Fig. 4. Panels and panel labels follow the same arrangement as in Extended Data Fig. 4. For convenience, lineage

labels are included in the upper left of each panel. Colors correspond to those in Extended Data Fig. 2. Error bar represents the control's 95% confidence interval across gels (mean $\pm$ 2 SEM; $n = 8$, Extended Data Fig. 4a–d).

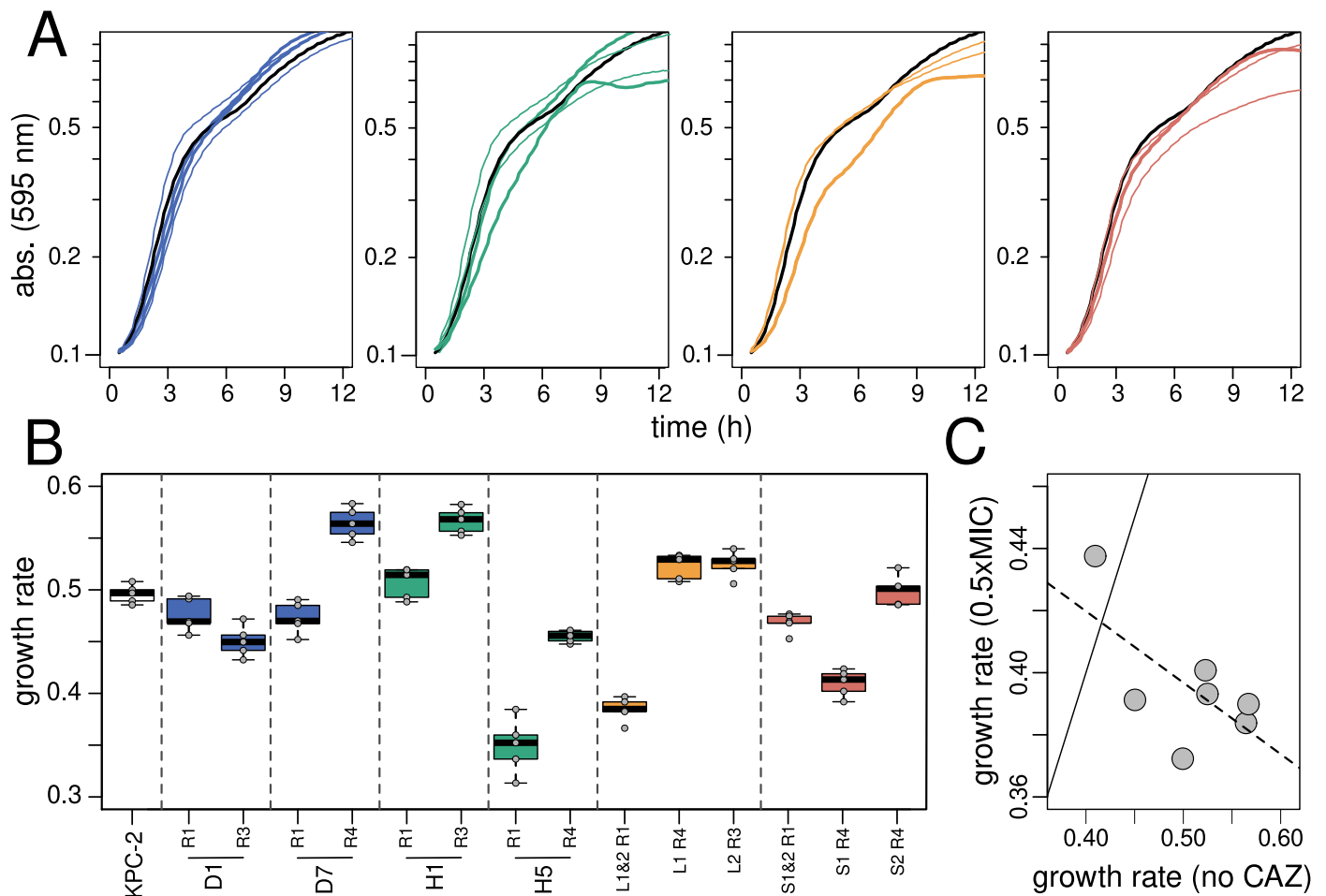


Extended Data Fig. 6 | Marked trade-offs between catalytic efficiency and periplasmic stability. (a) Catalytic efficiency shows a weak, positive, nonsignificant correlation with MIC increases (Pearson's $r = 0.37$, $P = 0.47$). (b) Periplasmic stability exhibits a strong negative correlation with MIC increases (Pearson's $r = -0.93$, $P = 0.007$). (c) Combining catalytic efficiency and periplasmic stability (as in ref. 83) still strongly negatively correlates with MIC

increases (Pearson's $r = -0.91$, $P = 0.0109$). Although higher stability should intuitively increase MIC by providing greater enzyme availability, this expectation only holds true in the absence of the activity-stability trade-offs observed here. Note these factors do not offset each other effectively because losses in stability are orders of magnitude greater than gains in catalytic efficiency.



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Extended Data Fig. 8 | Growth rates along evolutionary trajectories in the absence of antibiotic. (a) Optical density changes over 12 h of growth. The black line represents the strain with wild-type KPC-2, thick colored lines represent the first round strains, and the thinner colored lines represent plateau strains. (b) Boxplot summarizing maximum growth rates from the growth curves above. X-axis labels, excluding the control strain, indicate the evolutionary line (as in Table S2) and the round from which each isolate was taken (that is, rounds 1–4,

labeled R1–R4). Following R's default convention, boxes span the first (Q1) to third (Q3) quartiles, central lines mark the median, and whiskers reach the most extreme points within 1.5 times the interquartile range. (c) Growth rates in the absence of antibiotic show a moderate negative correlation with those measured in the presence of subinhibitory concentrations of CAZ, albeit non-significant ($R = -0.65$, $P = 0.11$). Solid line indicates the ideal one-to-one correspondence.

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Methods

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☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Antibodies

Antibodies used	Rat anti-HA high affinity monoclonal antibodies (Sigma-aldrich, Missouri, United States), Goat anti-rat immunoglobulin Horseradish Peroxidase (HRP) (Invitrogen)
Validation	The rat anti-HA high-affinity monoclonal antibody (Sigma-Aldrich) has been validated for Western blot applications by the manufacturer and has been widely used in the literature for detecting HA-tagged proteins. The goat anti-rat IgG HRP-conjugated antibody (Invitrogen) is a well-established secondary antibody for immunodetection in Western blotting, with validation provided by the manufacturer and supported by its extensive use in published studies.

Plants

Seed stocks	No plants were involved in this study.
Novel plant genotypes	No plants were involved in this study.
Authentication	No plants were involved in this study.